

Advanced (Carolina Reaper)

Q1. If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $(f + g)(x)$?

- (A) $x^2 + 2x + 1$
- (B) $x^2 - 2x - 1$
- (C) $x^2 + 2x - 1$
- (D) $x^2 - 2x + 1$
- (E) $x^2 + 1$

Q2. What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-2}$ when combined as $(f \cdot g)(x)$?

- (A) $(-\infty, 0) \cup (0, 2) \cup (2, \infty)$
- (B) $(-\infty, 0) \cup (0, \infty)$
- (C) $(-\infty, 2) \cup (2, \infty)$
- (D) $(-\infty, 0) \cup (2, \infty)$
- (E) $(-\infty, \infty)$

Q3. Simplify $(2f)(x)$ if $f(x) = x^2 - 3x + 1$.

- (A) $2x^2 - 6x + 2$
- (B) $2x^2 - 3x + 1$
- (C) $2x^2 + 6x - 2$
- (D) $x^2 - 3x + 1$
- (E) $2x^2 - 3x - 1$

Q4. Given $f(x) = x + 2$ and $g(x) = x - 2$, what is $(f - g)(x)$?

- (A) 4
- (B) $2x$
- (C) -4
- (D) $x + 4$
- (E) $x - 4$

Q5. If $h(x) = \frac{f(x)}{g(x)}$ where $f(x) = 2x + 1$ and $g(x) = x + 1$, what is $h(x)$?

- (A) 2
- (B) $\frac{2x+1}{x+1}$
- (C) $\frac{x+1}{2x+1}$
- (D) $2x + 1$
- (E) $x + 1$

Q6. What is the domain of $(f \div g)(x)$ where $f(x) = x^2 - 4$ and $g(x) = x - 2$?

- (A) $(-\infty, 2) \cup (2, \infty)$
- (B) $(-\infty, -2) \cup (2, \infty)$
- (C) $(-2, 2)$
- (D) $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$
- (E) $(-\infty, \infty)$

Q7. If $m(x) = x^2$ and $n(x) = 3x - 1$, what is $(m \cdot n)(x)$?

- (A) $3x^3 - x^2$
- (B) $3x^2 - x$
- (C) $3x^3 - x$
- (D) $3x^2 - x^3$
- (E) $x^2 + 3x - 1$

Q8. Given $f(x) = x^2 + 1$ and $g(x) = x - 1$, what is $(f + g)(2)$?

- (A) $4 + 1$
- (B) $2^2 + 1 + 2 - 1$
- (C) $2^2 + 2 - 1$
- (D) $2^2 - 2 + 1$
- (E) 5

Open Ended Questions (FRQ)

Q9. Given $f(x) = x - 2$ and $g(x) = x + 3$, determine the domain of $(f \cdot g)(x)$ and simplify the expression.

Q10. If $h(x) = \frac{f(x)}{g(x)}$ where $f(x) = 2x^2 - 3x + 1$ and $g(x) = x^2 + 1$, find $h(x)$ and state any restrictions on its domain.

Chef's Tips

- Ensure students have a clear understanding of function operations
- Remind students to simplify expressions after performing operations
- Emphasize the importance of identifying the domain of combined functions